

Lane2 Language Specification

Version: v1 draft

Status: working specification

Contents

1	Introduction	2
1.1	Scope	2
1.2	Compatibility	2
1.3	Experimental Features	2
1.4	Feedback	2
1.5	Reference	2
2	Syntax and Grammar	3
2.1	Notation	3
2.2	Lexical Grammar	3
2.3	Syntax Grammar	5
2.4	Comments	9
3	Type System	9
3.1	Type Objects	9
3.2	Type Well-Formedness	10
3.3	Type Equality	10
3.4	Type Inference	10
3.5	Static Semantics	11
4	Builtins	12
4.1	Required Ininsics	12
5	Prelude	12
5.1	Standard Operation Structs	12
5.2	Standard Preopen Values	13
6	Declarations	14
6.1	Top-Level Declarations	14
6.2	Struct Declarations	14
6.3	Enum Declarations	14
6.4	Function Declarations	14
6.5	Let Declarations	15
7	Scopes and Bindings	15
7.1	Namespaces	15
7.2	Local Bindings	16
7.3	Open Scope Extensions	16
7.4	Preopen	17
8	Expressions	17
8.1	Blocks	17
8.2	Conditional Expressions	18
8.3	Calls, Field Access, and Pipeline	18
9	Structs, Enums, and Open	18
9.1	Enums	19
9.2	Openable Struct Values	20
9.3	Struct Field Forwarding	20
10	Pattern Matching	20
11	Operators	21
12	Evaluation	22
13	Undefined Behavior	22
14	Conformance	22
15	Appendix	22

1 Introduction

Lane2 is a strict, pure, expression-oriented functional programming language. Its first implementation target is a parser, a type checker, and an AST interpreter; later implementations may add bytecode compilation, a virtual machine, a linker, and effect handling.

Lane2 is intentionally small. It has no mutable state, assignment, trait system, method dispatch, module system, or implicit typeclass-style instance search in v1. Instead, v1 is centered on nominal data, first-class functions, bidirectional local type inference, exhaustive pattern matching, and explicit operation values opened into lexical scope.

This document specifies Lane2/Core v1: the platform-independent part of Lane2 that should behave the same across the initial AST interpreter and later execution backends.

1.1 Scope

Lane2/Core v1 covers:

- source structure and declarations;
- lexical scope and name resolution;
- nominal struct and enum types;
- function and block expressions;
- local type inference;
- pattern matching;
- open, preopen, and operation-based operators;
- unsafe builtin expressions.

The following are outside Lane2/Core v1:

- mutation and assignment;
- modules and imports;
- bytecode and virtual machine semantics;
- linker entrypoint selection;
- algebraic effects;
- type aliases;
- traits, typeclasses, and interfaces;
- tuples and collection syntax.

1.2 Compatibility

This specification is a v1 draft. No compatibility is promised between draft revisions. Language rules may be added, removed, or changed while the first implementation is being developed.

Once a v1 implementation is declared stable, this section should be replaced by an explicit compatibility policy.

1.3 Experimental Features

This draft does not distinguish stable and experimental language features. Every rule in this document is provisional until v1 is stabilized.

Future drafts may mark individual features as experimental when their surface syntax or semantics are intentionally unsettled.

1.4 Feedback

Design feedback is tracked in the Lane2 repository. Implementation changes should update this specification, docs/language-v1.md, docs/prelude-v1.md, CONTEXT.md, and ADRs when the change affects user-visible semantics.

1.5 Reference

When referencing this draft, use:

> Lane2 Language Specification : Lane2/Core v1 draft.

2 Syntax and Grammar

2.1 Notation

```
grammar      ::= { production }
production   ::= nonTerminal "[:=" grammarExpression
grammarExpression ::=
    grammarSequence { "|" grammarSequence }
grammarSequence ::=
    { grammarItem }
grammarItem ::=
    terminal
    | nonTerminal
    | "[" grammarExpression "]"
    | "{" grammarExpression "}"
    | "(" grammarExpression ")"
    | grammarItem "?"
    | grammarItem "*"
    | grammarItem "+"

terminal     ::= "" terminalText ""
nonTerminal  ::= lowerCamelIdentifier
```

empty sequence denotes epsilon.

"A B" denotes sequencing.

"A | B" denotes choice.

"[A]" and "A?" denote optional occurrence.

"{ A }" and "A*" denote zero or more occurrences.

"A+" denotes one or more occurrences.

Parenthesized grammar expressions group without producing syntax.

2.2 Lexical Grammar

```
lexicalInput ::=
    (trivia | token)* eof
```

```
trivia ::=
    whitespace
    | lineTerminator
    | comment
```

```
token ::=
    keyword
    | reservedWord
    | identifier
    | intLiteral
    | stringLiteral
    | boolLiteral
    | operatorToken
    | punctuationToken
```

```
keyword ::=
    "struct"
    | "enum"
    | "fn"
    | "let"
    | "open"
    | "if"
    | "else"
    | "match"
    | "builtin"
```

```
reservedWord ::=
    "effect"
```

```

| "handler"
| "module"
| "import"
| "pub"
| "type"
| "trait"
| "interface"
| "mut"
| "return"

identifier ::=
  xidStart identifierContinue*
  | "_" identifierContinue+

identifierContinue ::=
  "_"
  | xidContinue
  | decimalDigit

boolLiteral ::=
  "true"
  | "false"

intLiteral ::=
  decimalDigit+

stringLiteral ::=
  "\"" stringElement* "\""

stringElement ::=
  asciiStringCharacter
  | stringEscape

asciiStringCharacter ::=
  asciiCodePointExceptControlBackslashQuote

stringEscape ::=
  "\\" " \" \"
  | "\\" " \" \"
  | "\\" "n"
  | "\\" "r"
  | "\\" "t"
  | "\\" "x" asciiHexByte

asciiHexByte ::=
  asciiHexLowByte
  | asciiHexHighByte

asciiHexLowByte ::=
  ("0" | "1" | "2" | "3" | "4" | "5" | "6") asciiHexDigit

asciiHexHighByte ::=
  "7" ("0" | "1" | "2" | "3" | "4" | "5" | "6" | "7")

operatorToken ::=
  "+"
  | "-"
  | "*"
  | "/"
  | "%"
  | "=="
  | "!="
  | "<"

```

```

| "<="
| ">"
| ">="
| "&&"
| "||"
| "!"
| "|>"

```

punctuationToken ::=

```

"("
| ")"
| "{"
| "}"
| "["
| "]"
| ","
| "."
| ":"
| "::"
| "->"
| "=>"
| "="
| "-"

```

whitespace ::=

```

U+0009
| U+000B
| U+000C
| U+0020

```

lineTerminator ::=

```

U+000A
| U+000D
| U+000D U+000A

```

decimalDigit ::=

```

"0" | "1" | "2" | "3" | "4" | "5" | "6" | "7" | "8" | "9"

```

asciiHexDigit ::=

```

decimalDigit
| "a" | "b" | "c" | "d" | "e" | "f"
| "A" | "B" | "C" | "D" | "E" | "F"

```

xidStart ::=

any Unicode code point with the XID_Start property

xidContinue ::=

any Unicode code point with the XID_Continue property

Lexical analysis uses maximal munch. If more than one token class matches the same maximal source span, keyword, reservedWord, and boolLiteral take priority over identifier.

2.3 Syntax Grammar

sourceFile ::=

```

topLevelDeclaration* eof

```

topLevelDeclaration ::=

```

structDeclaration
| enumDeclaration
| functionDeclaration
| topLevelLetDeclaration
| anonymousTopLevelLetDeclaration
| openDeclaration

```

```

structDeclaration ::=
    "struct" typeName typeParameters? "{" structMember* "}"

structMember ::=
    fieldDeclaration
  | fieldForwardingDeclaration

fieldDeclaration ::=
    fieldName space ":" space type

fieldForwardingDeclaration ::=
    "open" valueName

enumDeclaration ::=
    "enum" typeName typeParameters? "{" enumVariant* "}"

enumVariant ::=
    variantName enumPayload?

enumPayload ::=
    "(" commaSeparatedTypes? ")"

functionDeclaration ::=
    "fn" typeParameters? functionName "(" commaSeparatedParameters? ")" "->" type block

parameter ::=
    valueName space ":" space type

topLevelLetDeclaration ::=
    "let" valueName space ":" space type "=" expression

anonymousTopLevelLetDeclaration ::=
    "let" space ":" space type "=" expression

localLetDeclaration ::=
    "let" valueName typeAnnotation? "=" expression

typeAnnotation ::=
    space ":" space type

openDeclaration ::=
    "open" valueName

type ::=
    typeConstructor typeArguments?
  | functionType

typeConstructor ::=
    typeName

typeArguments ::=
    "[" commaSeparatedTypes "]"

typeParameters ::=
    "[" commaSeparatedTypeParameters "]"

functionType ::=
    typeParameters? "(" commaSeparatedTypes? ")" "->" type

expression ::=
    ifExpression
  | matchExpression

```

```

    | functionLiteral
    | block
    | pipelineExpression

pipelineExpression ::=
    binaryExpression { ">" pipelineRhs }

pipelineRhs ::=
    callExpression
    | functionLiteral

binaryExpression ::=
    unaryExpression { binaryOperator unaryExpression }

unaryExpression ::=
    unaryOperator unaryExpression
    | callExpression

callExpression ::=
    fieldExpression { callSuffix }

callSuffix ::=
    "(" commaSeparatedExpressions? ")"

fieldExpression ::=
    primaryExpression { "." fieldName }

primaryExpression ::=
    literal
    | valueName
    | qualifiedVariantExpression
    | structLiteral
    | builtinExpression
    | "(" expression ")"
    | "(" ")"

literal ::=
    intLiteral
    | stringLiteral
    | boolLiteral

qualifiedVariantExpression ::=
    typeName typeArguments? "::" variantName variantArguments?

variantArguments ::=
    "(" commaSeparatedExpressions? ")"

structLiteral ::=
    typeName typeArguments? "::" "{" commaSeparatedStructLiteralFields "}"

structLiteralField ::=
    fieldName
    | fieldName ":" expression

builtinExpression ::=
    "builtin" "(" stringLiteral ")"

functionLiteral ::=
    "fn" typeParameters? "(" commaSeparatedFunctionLiteralParameters? ")"
functionReturnAnnotation? block

functionLiteralParameter ::=
    valueName

```

```

    | valueName space ":" space type

functionReturnAnnotation ::=
    "->" type

block ::=
    "{" localItem* expression "}"

localItem ::=
    localLetDeclaration
    | functionDeclaration
    | openDeclaration

ifExpression ::=
    "if" expression block "else" block

matchExpression ::=
    "match" expression "{" matchArm* "}"

matchArm ::=
    pattern ">=" expression

pattern ::=
    "_"
    | valueName
    | literal
    | qualifiedVariantPattern
    | structPattern

qualifiedVariantPattern ::=
    typeName "::" variantName patternArguments?

patternArguments ::=
    "(" commaSeparatedPatterns? ")"

structPattern ::=
    typeName "::" "{" commaSeparatedStructPatternFields "}"

structPatternField ::=
    fieldName
    | fieldName ":" pattern

binaryOperator ::=
    "+" | "-" | "*" | "/" | "%"
    | "==" | "!=" | "<" | "<=" | ">" | ">="
    | "&&" | "||"

unaryOperator ::=
    "-" | "!"

commaSeparatedTypes ::=
    type ("," type)* ","?

commaSeparatedTypeParameters ::=
    typeParameter ("," typeParameter)* ","?

commaSeparatedParameters ::=
    parameter ("," parameter)* ","?

commaSeparatedExpressions ::=
    expression ("," expression)* ","?

commaSeparatedFunctionLiteralParameters ::=

```

```

functionLiteralParameter ("," functionLiteralParameter)* ",""?

commaSeparatedStructLiteralFields ::=
    structLiteralField ("," structLiteralField)* ",""?

commaSeparatedPatterns ::=
    pattern ("," pattern)* ",""?

commaSeparatedStructPatternFields ::=
    structPatternField ("," structPatternField)* ",""?

typeName ::=
    identifier

functionName ::=
    identifier

valueName ::=
    identifier

fieldName ::=
    identifier

variantName ::=
    identifier

typeParameter ::=
    identifier

space ::=
    whitespace+

```

The precedence and associativity of `binaryOperator`, `unaryOperator`, calls, field access, and pipeline expressions are defined in “Operators”.

2.4 Comments

```

comment ::=
    lineComment
  | blockComment

lineComment ::=
    "/" "/" lineCommentCharacter* lineTerminator?

lineCommentCharacter ::=
    any Unicode code point except U+000A or U+000D

blockComment ::=
    "/" "*" blockCommentCharacter* "*" "/"

blockCommentCharacter ::=
    any Unicode code point sequence that does not start "*"

```

Comments are trivia. Comments do not appear in the syntactic grammar.

3 Type System

3.1 Type Objects

The primitive type objects of Lane2/Core v1 are `Int`, `Bool`, `String`, and `Unit`.

`Int` is a signed 64-bit integer type.

`Bool` has the values `true` and `false`.

`String` is an ASCII string type.

`Unit` has the value `()`.

A `struct` declaration introduces a nominal struct type constructor.

An `enum` declaration introduces a nominal enum type constructor and its variant constructors.

A type parameter introduces a type variable scoped by its enclosing generic function, generic function literal, struct declaration, enum declaration, or generic function type.

A nominal type application consists of a type constructor and zero or more type arguments.

A function type consists of an ordered positional parameter list and a result type. Function types are not curried.

A generic function type consists of a type-parameter list followed by a function type.

Lane2/Core v1 has no tuple types, collection types, type aliases, trait constraints, typeclass constraints, interface constraints, or structural record types.

3.2 Type Well-Formedness

Type well-formedness is written as:

$$\Delta\Theta \vdash T \text{ type}$$

Δ is the type-constructor environment. It contains primitive type constructors and type constructors introduced by visible `struct` and `enum` declarations.

Θ is the type-variable environment. It contains type variables introduced by visible type-parameter binders.

A primitive type is well-formed.

A type variable is well-formed only when it is present in Θ .

A nominal type application is well-formed only when its type constructor resolves in Δ , the number of type arguments matches the type constructor arity, and all type arguments are well-formed.

A function type is well-formed only when every parameter type and the result type are well-formed.

A generic function type is well-formed only when its type parameters bind the type variables used by the following function type and that function type is well-formed under the extended type-variable environment.

3.3 Type Equality

Type equality is written as:

$$T \equiv U$$

Primitive type equality compares primitive type identity.

Type-variable equality compares type-variable binder identity.

Nominal type equality compares nominal type-constructor identity and then compares type arguments pairwise.

Function type equality compares parameter-list length, parameter types pairwise, and result type.

Distinct nominal type constructors are not equal merely because their declarations have the same fields or variants.

3.4 Type Inference

Lane2 uses bidirectional local type inference.

$$\Gamma \vdash e \Rightarrow T$$

$$\Gamma \vdash e \Leftarrow T$$

$$\Gamma \vdash p \Leftarrow T \Rightarrow \Gamma_p$$

$$\Gamma \vdash e \Rightarrow T$$

means expression e synthesizes type T .

$$\Gamma \vdash e \Leftarrow T$$

means expression e checks against expected type T .

$$\Gamma \vdash p \Leftarrow T \Rightarrow \Gamma_p$$

means pattern p checks against scrutinee type T and produces pattern bindings Γ_p .

Type information may synthesize upward from an expression or check downward from an expected type.

The type checker does not create global Hindley-Milner constraints. A binding's type is not inferred from later uses of the bound name.

Function literals may omit parameter types only when checked against an expected function type. Without an expected function type, every value parameter of a function literal must have an explicit type annotation.

Generic function literals without an expected generic function type must explicitly declare type parameters and explicitly annotate value parameters.

Generic function calls and generic data constructors may instantiate type parameters at the use site when the use site is unambiguous.

3.5 Static Semantics

The static semantics of Lane2/Core v1 use these judgments.

$$\Delta\Theta \vdash T \text{ type}$$

$$T \equiv U$$

$$\Gamma \vdash e \Rightarrow T$$

$$\Gamma \vdash e \Leftarrow T$$

$$\Gamma \vdash p \Leftarrow T \Rightarrow \Gamma_p$$

$$\Gamma \vdash \text{sourceFile} \text{ ok}$$

Γ is the value environment for the current checking position. Name resolution, ordered top-level value scope, open scope extensions, and preopen are specified in “Scopes and Bindings”.

$$\Gamma \vdash \text{sourceFile} \text{ ok}$$

means the source file is syntactically valid, all referenced types are well-formed, all names resolve without ambiguity, all expressions type-check, and all match expressions are exhaustive.

A Lane2/Core v1 source file is statically valid only if:

1. every type annotation denotes a well-formed type;
2. every expression synthesizes or checks against its required type;
3. every local binding obeys sequential local scope;
4. every top-level value and top-level open obeys ordered value scope;
5. every function body checks under the source file's top-level environment;

6. every match expression is exhaustive;
7. every unresolved name, ambiguous name, ill-formed type, or type mismatch is rejected.

4 Builtins

`builtin("...")` is an unsafe expression.

The checker does not interpret intrinsic strings. Instead, `builtin` receives its type from direct expected context:

```
fn int_add(a : Int, b : Int) -> Int {
  builtin("%i64_add")
}
```

This is valid because the function return type provides the expected type `Int` for the body expression.

This is invalid:

```
let x = builtin("%anything")
```

There is no direct expected type.

Incorrect builtin use can produce undefined behavior. Lane2's type safety guarantee applies only to programs that do not misuse builtin.

4.1 Required Ininsics

A conforming Lane2/Core v1 implementation provides these portable intrinsic names:

Intrinsic	Expected type
<code>%i64_add</code>	<code>(Int, Int) -> Int</code>
<code>%i64_sub</code>	<code>(Int, Int) -> Int</code>
<code>%i64_mul</code>	<code>(Int, Int) -> Int</code>
<code>%i64_div</code>	<code>(Int, Int) -> Int</code>
<code>%i64_rem</code>	<code>(Int, Int) -> Int</code>
<code>%i64_neg</code>	<code>(Int) -> Int</code>
<code>%i64_equal</code>	<code>(Int, Int) -> Bool</code>
<code>%i64_less</code>	<code>(Int, Int) -> Bool</code>
<code>%string_equal</code>	<code>(String, String) -> Bool</code>

Other intrinsic names are implementation-defined unsafe builtins.

`%bool_and`, `%bool_or`, `%bool_not`, and `%bool_equal` are not required intrinsics. The standard prelude defines boolean operations as ordinary Lane2 functions and anonymous operation values using `if`; `&&` and `||` supply their right operands as thunks.

5 Prelude

The standard prelude is implementation-supplied Lane2 source checked before user code. It is not a module system.

Prelude declarations provide standard operation structs, primitive wrappers around required intrinsics, derived primitive operations written in Lane2, and anonymous top-level values that populate the initial preopen namespace.

Prelude entries are ordinary Lane2 values and types except where the compiler recognizes operation field pairs for operator aliases.

5.1 Standard Operation Structs

Arithmetic operations:

```
struct Add[T] {
  add : (T, T) -> T
}
```

```
struct Sub[T] {  
  sub : (T, T) -> T  
}
```

```
struct Mul[T] {  
  mul : (T, T) -> T  
}
```

```
struct Div[T] {  
  div : (T, T) -> T  
}
```

```
struct Rem[T] {  
  rem : (T, T) -> T  
}
```

```
struct Neg[T] {  
  neg : (T) -> T  
}
```

Boolean operations:

```
struct And {  
  and : (Bool, () -> Bool) -> Bool  
}
```

```
struct Or {  
  or : (Bool, () -> Bool) -> Bool  
}
```

```
struct Not {  
  not : (Bool) -> Bool  
}
```

Equality and ordering operations:

```
struct Equal[T] {  
  equal : (T, T) -> Bool  
  not_equal : (T, T) -> Bool  
}
```

```
struct Compare[T] {  
  equal_impl : Equal[T]  
  open equal_impl  
  less : (T, T) -> Bool  
  less_eq : (T, T) -> Bool  
  greater : (T, T) -> Bool  
  greater_eq : (T, T) -> Bool  
}
```

Operation laws are API conventions, not compiler-checked rules.

5.2 Standard Preopen Values

The prelude may populate the initial preopen namespace with anonymous top-level values. For example, primitive arithmetic and boolean operations can be exposed as follows:

```
let : Add[Int] = Add::{ add: int_add }  
let : Sub[Int] = Sub::{ sub: int_sub }  
let : Mul[Int] = Mul::{ mul: int_mul }  
let : Div[Int] = Div::{ div: int_div }  
let : Rem[Int] = Rem::{ rem: int_rem }  
let : Neg[Int] = Neg::{ neg: int_neg }
```

```
let int_equal_ops : Equal[Int] = make_equal(int_equal)
let : Compare[Int] = make_compare(int_equal_ops, int_less)
```

```
let : And = And::{ and: bool_and }
let : Or = Or::{ or: bool_or }
let : Not = Not::{ not: bool_not }
let : Equal[Bool] = Equal::{
  equal: fn(a : Bool, b : Bool) {
    if a {
      b
    } else {
      bool_not(b)
    }
  },
  not_equal: fn(a : Bool, b : Bool) {
    if a {
      bool_not(b)
    } else {
      b
    }
  },
}
```

```
let : Equal[String] = make_equal(string_equal)
```

Boolean conjunction, disjunction, negation, and equality do not require boolean intrinsics; they can be defined with ordinary if expressions in the prelude.

6 Declarations

Declarations introduce types, functions, values, and open scope extensions.

6.1 Top-Level Declarations

Top-level declarations are described by the `topLevelDefinition` grammar in “Syntax and Grammar”.

Top-level forms include struct declarations, enum declarations, named function declarations, typed value declarations, anonymous typed value declarations, and open declarations.

6.2 Struct Declarations

Struct declarations use the grammar in “Structs, Enums, and Open”.

6.3 Enum Declarations

Enum declarations use the grammar in “Structs, Enums, and Open”.

6.4 Function Declarations

Functions are uncurried. There is no automatic currying or partial application.

Function definitions use block bodies:

```
functionDeclaration:
  'fn' [ typeParameters ] functionName '(' [ parameter { ',' parameter } [ ',' ] ] ')' '->'
  type block
```

```
parameter:
  valueName ':' type
```

```
functionLiteral:
  'fn' [ typeParameters ] '(' [ functionLiteralParameter { ',' functionLiteralParameter }
  [ ',' ] ] ')' [ '->' type ] block
```

```
functionLiteralParameter:
  valueName
  | valueName ':' type
```

```
fn add(a : Int, b : Int) -> Int {
  a + b
}
```

There is no `=` expr function body form.

Function result types use `->`, not `::`.

All named functions must state every parameter type and the result type.

Generic named functions place type parameters after `fn` and before the name:

```
fn[A] id(value : A) -> A {
  value
}
```

Function parameters are positional in v1. Labeled function parameters are not supported.

Function literals produce function values:

```
let f : (Int, Int) -> Int = fn(a, b) {
  a + b
}
```

Generic function literals place type parameters after `fn`:

```
let id = fn[A](value : A) {
  value
}
```

6.5 Let Declarations

Named top-level `let` declarations must include type annotations. Anonymous top-level `let` declarations must include type annotations and must have a struct type.

Local `let` declarations may omit type annotations when their initializer can synthesize a type by local type inference.

7 Scopes and Bindings

Top-level type and function definitions form recursive definition groups.

Top-level `struct`, `enum`, and `fn` definitions may refer to each other regardless of textual order.

Top-level value definitions and top-level open declarations follow ordered value scope:

- a top-level `let` initializer may refer only to values already available at that declaration point;
- a top-level `open` may open only a value already available at that declaration point;
- a top-level function body may refer to any top-level value or function, even one declared later.

Example:

```
fn read_x() -> Int {
  x
}
```

```
let x : Int = 1
```

This is valid because the function body sees the complete top-level environment.

```
let y : Int = x
let x : Int = 1
```

This is invalid because top-level value initializers follow ordered value scope.

7.1 Namespaces

Lane2 has separate type and value namespaces.

Type and value names may use the same spelling:

```
enum Option[A] {
  none
  some(A)
}
```

```
let Option : Int = 42
```

Syntax of the form `Type::member` resolves `Type` in the type namespace.

```
let x : Option[Int] = Option::some(1)
```

The `Option` on the left of `::` denotes the enum type, not the value named `Option`.

7.2 Local Bindings

Local `let` bindings are sequential. A local binding is visible only to later items and the final expression in the same block.

Local `let` bindings may omit type annotations when their initializer can synthesize a type by local type inference:

```
let x = 1
```

Local named functions are also sequential. They may call themselves, but local mutually recursive groups and forward references are not supported:

```
fn f(n : Int) -> Int {
  fn loop(x : Int) -> Int {
    if x == 0 {
      0
    } else {
      loop(x - 1)
    }
  }
  loop(n)
}
```

Local value names may shadow earlier local value names.

7.3 Open Scope Extensions

`open value` opens a struct value.

The operand of `open` must be a visible value name with a struct type. It cannot be an arbitrary expression:

```
open ops
```

This is invalid:

```
open make_ops(10)
```

An open scope extension exposes the struct field values as unqualified names from the declaration point to the end of the current lexical scope.

At top level, `open` extends to the end of the file:

```
open int_add_ops
```

```
fn add_one(x : Int) -> Int {
  x + 1
}
```

Inside blocks, `open` is a local item and must appear before the final expression:

```
{
  open int_add_ops
  x + y
}
```

Local resolution uses the nearest preceding local binding or open scope extension, then falls back to preopen.

7.4 Preopen

The preopen namespace is a default-open namespace populated by anonymous top-level values.

```
let : Add[Int] = Add::{ add: int_add }
```

An anonymous top-level value must have a struct type. It exposes field values, not generated accessors.

Prelude-provided anonymous values are checked before user code, so their preopen entries are visible throughout user code.

User-defined anonymous top-level values extend preopen from their declaration point to the end of the top-level scope.

Preopen conflicts are errors.

Top-level open belongs to the global layer and conflicts with ordinary top-level names or preopened names.

Local open may shadow preopen inside its lexical scope.

8 Expressions

Expressions compute values. Lane2 v1 is expression-oriented and has no expression statements.

8.1 Blocks

A block expression contains local items followed by one final expression.

```
block:
  '{' { localItem } expression '}'

localItem:
  localLet
  | localFunctionDeclaration
  | openDeclaration

localLet:
  'let' valueName [ ':' type ] '=' expression

topLevelLet:
  'let' valueName ':' type '=' expression

anonymousTopLevelLet:
  'let' ':' type '=' expression

openDeclaration:
  'open' valueName

{
  let x = 1
  fn double(n : Int) -> Int {
    n + n
  }
  double(x)
}
```

Allowed local items:

- let,
- named fn,
- open.

Local type definitions are not supported in v1.

A block does not contain expression statements. This is invalid:

```
{
  f(x)
```

```
    g(y)
}
```

An empty block is invalid. Write `()` explicitly when a `Unit` value is required.

8.2 Conditional Expressions

`if` is an expression:

```
if condition {
  then_value
} else {
  else_value
}
```

The `else` branch is mandatory.

Both branches must have the same type.

8.3 Calls, Field Access, and Pipeline

Function call:

```
f(x, y)
```

Field access:

```
ops.add
```

Field access followed by call:

```
ops.add(x, y)
```

This is not method call syntax. `Lane2 v1` has no method receiver lookup.

Pipeline syntax is supported:

```
value |> f(a, b)
```

It rewrites to:

```
f(value, a, b)
```

The right-hand side of `|>` must be a call or function literal.

The following are invalid:

```
value |> f
value |> f(_, y)
```

Placeholders are not supported.

9 Structs, Enums, and Open

`structDeclaration`:

```
'struct' typeName [ typeParameters ] '{' { structMember } '}'
```

`structMember`:

```
  fieldDeclaration
| fieldForwarding
```

`fieldDeclaration`:

```
  fieldName ':' type
```

`fieldForwarding`:

```
'open' fieldName
```

`structLiteral`:

```
  typeName [ typeArguments ] '::' '{' structLiteralField { ',' structLiteralField } [ ',' ]
  '}'
```

```
structLiteralField:
    fieldName
  | fieldName ':' expression
```

Struct literals are qualified:

```
let p : Point = Point::{ x: 1, y: 2 }
```

Struct field punning is allowed:

```
let p : Point = Point::{ x, y }
```

This is equivalent to:

```
let p : Point = Point::{ x: x, y: y }
```

Struct literals must provide every field exactly once.

The following are not supported:

- anonymous record literals,
- default fields,
- spread,
- copy update,
- field update.

Fields are accessed with dot syntax:

```
p.x
```

V1 has no field visibility modifiers. Every struct field is accessible wherever the struct value is accessible.

9.1 Enums

```
enumDeclaration:
    'enum' typeName [ typeParameters ] '{' { enumVariant } '}'
```

```
enumVariant:
    variantName [ '(' [ type { ',' type } [ ',' ] ] ')' ]
```

```
qualifiedVariantExpression:
    typeName [ typeArguments ] '::' variantName [ '(' [ expression { ',' expression } [ ',' ] ]
    ')' ]
```

```
unqualifiedVariantExpression:
    variantName [ '(' [ expression { ',' expression } [ ',' ] ] ')' ]
```

Enum variants are scoped to their enum.

Variant names may be lowercase or uppercase. Capitalization has no semantic role.

Payloadless variants are values and are written without call parentheses:

```
let x : Option[Int] = Option::none
```

This is invalid:

```
Option::none()
```

Variant payloads are positional:

```
enum Expr {
    int(Int)
    add(Expr, Expr)
}
```

Labeled variant payloads are not part of v1. Named product data must be represented with a struct:

```
struct Node[A] {
    left : Tree[A]
    value : A
}
```

```

    right : Tree[A]
  }

```

```

enum Tree[A] {
  leaf(A)
  node(Node[A])
}

```

In expressions, unqualified variants are allowed when unambiguous:

```
let x : Option[Int] = some(1)
```

If more than one visible enum has a variant named `some`, the unqualified expression is invalid unless another directly local rule disambiguates it. A qualified variant is always allowed:

```
let x : Option[Int] = Option::some(1)
```

In patterns, variants must always be qualified.

9.2 Openable Struct Values

Only struct values can be opened.

Opening a struct value exposes its field values as unqualified bindings according to the scope rules in “Scopes and Bindings”.

Enum values, function values, primitive values, and arbitrary expressions are not openable.

9.3 Struct Field Forwarding

A struct declaration may contain `open` field entries:

```

struct Compare[T] {
  equal_impl : Equal[T]
  open equal_impl
  less : (T, T) -> Bool
  less_eq : (T, T) -> Bool
  greater : (T, T) -> Bool
  greater_eq : (T, T) -> Bool
}

```

When a value of this struct type is opened, forwarded fields are exposed too.

Struct field forwarding affects only what is exposed by opening the containing struct value. It does not open the field inside ordinary function bodies.

10 Pattern Matching

`match` is an expression:

`matchExpression`:

```
'match' expression '{' { matchArm } '}'
```

`matchArm`:

```
pattern '=>' expression
```

`pattern`:

```

'_'
| valueName
| literal
| qualifiedVariantPattern
| structPattern

```

`qualifiedVariantPattern`:

```
typeName '::' variantName [ '(' [ pattern { ',' pattern } [ ',' ] ] ')']
```

`structPattern`:

```
typeName '::' '{' structPatternField { ',' structPatternField } [ ',' ] '}'
```

```

structPatternField:
  fieldName
  | fieldName ':' pattern
match value {
  Option::none => fallback
  Option::some(x) => x
}

```

Match arms use `pattern => expression`.

Every match must be exhaustive.

V1 patterns:

- wildcard pattern: `_`,
- variable binding pattern: `name`,
- literal pattern,
- qualified enum variant pattern,
- qualified struct pattern.

Enum variants in patterns must be qualified:

```

Option::some(x)
Option::none

```

Bare identifiers in patterns are always variable bindings:

```

match value {
  x => x
}

```

Struct patterns use qualified syntax and must list all fields:

```

match p {
  Point::{ x, y } => x + y
}

```

Struct pattern punning is allowed. Explicit renaming is allowed:

```

match p {
  Point::{ x: a, y: b } => a + b
}

```

Rest patterns, spread patterns, guards, or-patterns, as-patterns, and `is` pattern expressions are not supported in v1.

11 Operators

Operators are aliases for recognized prelude operation fields.

There is no type-directed instance search. An operator is available only when the relevant operation field is available through `preopen` or an open scope extension.

Primitive operators are not special-cased. For example, `1 + 2` requires an available `Add::add` operation for `Int`.

Recognized operator mappings:

Operator	Operation
<code>+</code>	<code>Add::add</code>
<code>-</code>	<code>Sub::sub</code>
<code>*</code>	<code>Mul::mul</code>
<code>/</code>	<code>Div::div</code>
<code>%</code>	<code>Rem::rem</code>

unary -	Neg::neg
==	Equal::equal
!=	Equal::not_equal
<	Compare::less
<=	Compare::less_eq
>	Compare::greater
>=	Compare::greater_eq
&&	And::and with a thunked right operand
	Or::or with a thunked right operand
!	Not::not

&& and || are short-circuit boolean operators. They are recognized mappings to And::and and Or::or, but the right operand is passed as a zero-argument function instead of being evaluated before the operation call.

a && b desugars to a call equivalent to and(a, fn() { b }) after resolving an available And::and operation. a || b follows the same rule with Or::or. Both operators are defined only for Bool.

Concrete expression precedence, associativity, and unary/binary disambiguation follow the MoonBit parser reference where Lane2 has not deliberately removed a feature.

12 Evaluation

Lane2 v1 is pure and strict.

- Evaluating a safe expression depends only on its inputs and produces a value.
- Function arguments are evaluated before the function body runs.
- if and match evaluate only the selected branch.
- There is no mutable binding, mutable field, assignment, field update, or implicit statement sequencing.
- IO and other effects are not part of v1 core semantics.

The Unit value is written explicitly as ().

Empty blocks are invalid. A block must contain exactly one final expression after any local items.

Function calls evaluate all arguments before entering the function body. The only v1 operator forms that delay a subexpression are && and ||, which pass the right operand as a thunk to the resolved prelude operation.

13 Undefined Behavior

Incorrect builtin use can produce undefined behavior. Lane2’s type safety guarantee applies only to programs that do not misuse builtin.

Invalid integer arithmetic is undefined behavior in v1. This includes signed 64-bit overflow, division by zero, remainder by zero, MIN_INT / -1, and negating MIN_INT.

14 Conformance

The words “must”, “must not”, “may”, and “should” are normative in this document.

Text marked as rationale, examples, or notes is informative unless it explicitly uses normative language.

An implementation conforms to this draft if it accepts all valid programs described here, rejects invalid programs described here, and gives the specified typing and evaluation behavior for safe programs.

Programs that misuse builtin are outside Lane2’s safety guarantee.

15 Appendix

15.1 Deliberately Omitted From V1

V1 omits:

- mutation and assignment,
- field update and copy update,
- modules and imports,
- local type definitions,
- labeled function parameters,
- labeled enum payloads,
- type aliases,
- traits, typeclasses, interfaces,
- method syntax,
- tuple types,
- collections,
- pattern guards,
- or-patterns,
- as-patterns,
- is pattern expressions,
- placeholder pipeline.